

1. NAME OF THE MEDICINAL PRODUCT

B. Braun 20% Glucose Intravenous Infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

1000 ml of solution for infusion contain:

Glucose monohydrate 220.0 g

(equivalent to glucose) (200.0 g)

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for infusion

Clear and colourless or faintly yellow solution, free from particles

Energy 3350 kJ/l $\pm$ 800 kcal/l

Theoretical osmolality 1110 mOsm/l

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

– Administration of glucose for caloric support

– Carbohydrate component in parenteral nutrition regimes

– Therapy of hypoglycaemia

4.2 Posology and method of administration

Posology

The dosage of the solution depends on the patient's individual glucose and fluid requirements.

Fluid balance, serum glucose, and other electrolytes may need to be monitored before and during administration, especially in patients with increased non-osmotic vasopressin release (syndrome of inappropriate antidiuretic hormone secretion, SIADH) and in patients co-medicated with vasopressin agonist drugs due to the risk of hyponatraemia.

Monitoring of serum sodium is particularly important for physiologically hypotonic fluids. Glucose 20% may become hypotonic after administration due to glucose metabolism in the body (see sections 4.4. 4.5 and 4.8).

Adults and adolescents from 15th year of life

The maximum daily dose is 30 ml per kg body weight per day, corresponding to 6 g of glucose per kg body weight per day.

The maximum infusion rate is 1.25 ml per kg body weight per hour, corresponding to 0.25 g of glucose per kg body weight per hour.

Thus, for a patient weighing 70 kg the maximum infusion rate is approximately 87 ml per hour, resulting in a glucose intake of 17.5 g per hour.

Paediatric population

The maximum daily dose, in g of glucose per kg body weight and in ml of solution per kg body weight per day, is for:

Pre-term neonates:	18 g per kg body weight	90 ml per kg body weight
Term neonates:	15 g per kg body weight	75 ml per kg body weight
1 st – 2 nd year:	15 g per kg body weight	75 ml per kg body weight
3 rd – 5 th year:	12 g per kg body weight	60 ml per kg body weight
6 th – 10 th year:	10 g per kg body weight	50 ml per kg body weight
11 th – 14 th year:	8 g per kg body weight	40 ml per kg body weight

For use in neonates, due account should be taken of the high osmolality of the solution (see section 3).

When administering this solution, the total daily fluid intake must be taken into account. The recommended daily parenteral fluid intake for children is as follows:

1 st day of life:	60 – 120 ml per kg body weight per day
2 nd day of life:	80 – 120 ml per kg body weight per day
3 rd day of life:	100 – 130 ml per kg body weight per day
4 th day of life:	120 – 150 ml per kg body weight per day
5 th day of life:	140 – 160 ml per kg body weight per day
6 th day of life:	140 – 180 ml per kg body weight per day
1 st month, prior to establishment of stable growth:	140 – 170 ml per kg body weight per day
1 st month, after establishment of stable growth:	140 – 160 ml per kg body weight per day
2 nd – 12 th month of life:	120 – 150 ml per kg body weight per day
2 nd year:	80 – 120 ml per kg body weight per day
3 rd – 5 th year:	80 – 100 ml per kg body weight per day
6 th – 12 th year:	60 – 80 ml per kg body weight per day
13 th – 18 th year:	50 – 70 ml per kg body weight per day

Elderly patients

Basically the same dosage as for adults applies, but caution should be exercised in patients suffering from further diseases like cardiac insufficiency or renal insufficiency that may frequently be associated with advanced age.

Patients with impaired glucose metabolism

If the oxidative metabolism of glucose is impaired (e.g. in the early post-operative or post-traumatic period or in the presence of hypoxia or organ failure), the dosage should be adjusted to keep the blood glucose level close to normal values. Close monitoring of blood glucose levels is recommended in order to prevent hyperglycaemia.

Method of administration

Intravenous use. For central venous infusion only.

4.3 Contraindications

– Hyperglycaemia, not responding to insulin doses of up to 6 units insulin/hour

– Delirium tremens if such patients are already dehydrated

– Acute states of shock and collapse

– Metabolic acidosis

Since the administration of glucose solutions is accompanied by the administration of free water, further contraindications may arise e.g.:

– Hyperhydration

– Pulmonary oedema

– Acute congestive heart failure

4.4 Special warnings and precautions for use

General

B. Braun 20% Glucose Intravenous Infusion is a hypertonic solution. In the body, however, glucose containing fluids can become extremely physiologically hypotonic due to rapid glucose metabolism (see section 4.2).

Depending on the tonicity of the solution, the volume and rate of infusion and depending on a patient's underlying clinical condition and capability to metabolize glucose, intravenous administration of glucose can cause electrolyte disturbances most importantly hypo- or hyperosmotic hyponatraemia.

Hyponatraemia:

Patients with non-osmotic vasopressin release (e.g. in acute illness, pain, post-operative stress, infections, burns, and CNS diseases), patients with heart-, liver- and kidney diseases and patients exposed to vasopressin agonists (see section 4.5) are at particular risk of acute hyponatraemia upon infusion of hypotonic fluids.

Acute hyponatraemia can lead to acute hyponatraemic encephalopathy (brain oedema) characterized by headache, nausea, seizures, lethargy and vomiting. Patients with brain oedema are at particular risk of severe, irreversible and life-threatening brain injury.

Children, women in the fertile age and patients with reduced cerebral compliance (e.g. meningitis, intracranial bleeding, and cerebral contusion) are at particular risk of the severe and life-threatening brain swelling caused by acute hyponatraemia.

Administration of glucose solutions is not recommended after acute ischaemic strokes as hyperglycaemia has been reported to worsen ischaemic brain damage and impair recovery.

Application of hyperosmolar glucose solutions in patients with damaged haematoencephalic barrier may lead to increase of intracranial/intraspinal pressure.

Glucose infusions should not be started before existing fluid and electrolyte deficiencies like hypotonic dehydration, hyponatraemia and hypokalaemia have adequately been corrected.

This solution should be used with caution in patients with

– Hypervolaemia

– Renal insufficiency

– Cardiac insufficiency

– Increased serum osmolality

– Known subclinical diabetes mellitus or carbohydrate intolerance for any reason.

Unstable metabolism (e.g. postoperatively or after injuries, hypoxia, organ insufficiencies) impairs oxidative metabolism of glucose and may lead to metabolic acidosis.

States of hyperglycaemia should be adequately monitored and treated with insulin. The application of insulin causes additional shifts of potassium into the cells and may therefore cause or increase hypokalaemia.

Sudden discontinuation of high glucose infusion rates can lead to profound hypoglycaemia due to the accompanying high serum insulin concentrations. This applies especially to children less than 2 years of age, patients with diabetes mellitus and patients with other disease states associated with impaired glucose homeostasis. In obvious cases, the glucose infusion should be tapered off within the last 30–60 minutes of the infusion. As a precaution it is recommended that each individual patient be monitored for 30 minutes for hypoglycaemia on the first day of abrupt discontinuation of parenteral nutrition.

Clinical monitoring should include blood glucose, serum electrolytes, fluid and acid-base balance in general. A focus should be put on the sodium level as glucose solutions provide free water to the body and may therefore cause or worsen hyponatraemia. Frequency and kind of laboratory testing depend on the overall condition of the patient, the prevailing metabolic situation, the administered dose and the duration of treatment. Also monitor total volume and amount of glucose administered.

Parenteral nutrition in malnourished or depleted patients with full doses and full infusion rates from the very beginning and without adequate supplementation of potassium, magnesium and phosphate may lead to the refeeding syndrome, characterized by hypokalaemia, hypophosphataemia and hypomagnesaemia. Clinical manifestations may develop within a few days of starting parenteral nutrition. In such patients, infusion regimens should be built up gradually. Adequate supplementation of electrolytes according to deviations from normal values is necessary.

Special attention should be paid to hypokalaemia. Then, supplementation of potassium is mandatory.

Electrolytes and vitamins must be supplied as necessary. Vitamin B, especially thiamine, is needed for glucose metabolism.

Glucose infusions should not be administered through the same infusion equipment, simultaneously before, or after administration of blood, because of the possibility of pseudo-agglutination.

It should be noted that this solution constitutes only one component of parenteral nutrition. In total parenteral nutrition, glucose infusions should always be combined with an adequate supply of amino acids, lipids, electrolytes, vitamins and trace elements.

Paediatric population

For treatment of hypoglycaemia in children, use of 10% glucose solution is recommended.

Children in the 1st and 2nd year of life are especially at risk for rebound hypoglycaemia after abrupt discontinuation of high infusion rates, see above.

4.5 Interaction with other medicinal products and other forms of interaction

Interactions with medicinal products with an influence on glucose metabolism should be considered.

Drugs leading to an increased vasopressin effect.

The below listed drugs increase the vasopressin effect, leading to reduced renal electrolyte free water excretion and increase the risk of hospital acquired hyponatraemia following inappropriately balanced treatment with i.v. fluids (see sections 4.2, 4.4 and 4.8).

• Drugs stimulating vasopressin release, e.g.: Chlorpropamide, clofibrate, carbamazepine, vincristine, selective serotonin reuptake inhibitors, 3,4-methylenedioxy-N-methamphetamine, ifosfamide, antipsychotics, narcotics

• Drugs potentiating vasopressin action, e.g.: Chlorpropamide, NSAIDs, cyclophosphamide

• Vasopressin analogues, e.g.: Desmopressin, oxytocin, vasopressin, terlipressin

Other medicinal products increasing the risk of hyponatraemia also include diuretics in general and antiepileptics such as oxcarbazepine.

4.6 Fertility, pregnancy and lactation

Pregnancy

There are no or limited data (less than 300 pregnancy outcomes) from the use of glucose monohydrate in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity (see section 5.3).

The use of B. Braun 20% Glucose Intravenous Infusion may be considered during pregnancy, if clinically needed.

B. Braun 20% Glucose Intravenous Infusion should be administrated with special caution for pregnant women during labour particularly if administered in combination with oxytocin due to the risk of hyponatraemia (see section 4.4, 4.5 and 4.8).

Careful monitoring of blood glucose is necessary.

Breast-feeding

Glucose/metabolites are excreted in human milk, but at therapeutic doses of B. Braun 20% Glucose Intravenous Infusion no effects on the breast-fed newborns/infants are anticipated. B. Braun 20% Glucose Intravenous Infusion can be used during breast-feeding as indicated.

Fertility

No special precautions.

Approval for Printing

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4.7 Effects on ability to drive and use machines

The solution has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable effects

General

Undesirable effects are listed according to their frequencies as follows:

Very common	(≥ 1/10)
Common	(≥ 1/100 to < 1/10)
Uncommon	(≥ 1/1,000 to < 1/100)
Rare	(≥ 1/10,000 to < 1/1,000)
Very rare	(< 1/10,000)
Not known	(cannot be estimated from the available data)

General disorders and administration site conditions:

Not known: Local reactions at the site of administration, including local pain, vein irritation, thrombophlebitis or tissue necrosis in case of extravasation.

Metabolism and nutrition disorders:

Not known: Hospital Acquired Hyponatraemia

Neurological disorders:

Not known: Hyponatraemic encephalopathy

Hospital acquired hyponatraemia may cause irreversible brain injury and death due to development of acute hyponatraemic encephalopathy (see sections 4.2 and 4.4).

4.9 Overdose

Symptoms

Symptoms of glucose overdose

Excessive glucose infusions can cause hyperglycaemia, glucosuria, hyperosmolar dehydration and in extreme case overdose can lead to hyperglycaemic-hyperosmolar coma. In cases of gross overdosing lipogenesis resulting in hepatic steatosis is possible.

Symptoms of fluid overdose

Fluid overdose may result in hyperhydration with increased skin tension, venous congestion, oedema – possibly also lung or brain oedema – dilution of serum electrolytes, electrolyte imbalances, notably hyponatraemia and hypokalaemia (see section 4.4), and acid-base imbalances.

Clinical symptoms of water intoxication may occur like nausea, vomiting and spasms.

Treatment

The primary therapeutic measure is dose reduction or cessation of infusion, depending on the severity of the symptoms. Disorders of the carbohydrate and electrolyte metabolism are treated by insulin administration and appropriate electrolyte substitution, respectively.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Solutions for parenteral nutrition, carbohydrates, ATC code: B05B A03

Pharmacodynamic effects

Glucose is metabolized ubiquitously as the natural substrate of the cells of the body. Under physiological conditions glucose is the most important energy-supplying carbohydrate with a caloric value of approx. 16.7 kJ/g or 4 kcal/g. In adults, the normal concentration of glucose in blood is reported to be 70 – 100 mg/dl or 3.9 to 5.6 mmol/l (fasting).

5.2 Pharmacokinetic properties

Absorption

Since the solution is administered intravenously, its bioavailability is 100%.

Distribution

After infusion, glucose is first distributed in the intravascular space and then is taken up into the intracellular space.

Biotransformation

In glycolysis, glucose is metabolized to pyruvate. Under aerobic conditions pyruvate is completely oxidized to carbon dioxide and water. In case of hypoxia, pyruvate is converted to lactate. Lactate can be partially reintroduced into the glucose metabolism (Cori cycle). Glucose utilisation disturbances (glucose intolerance) can occur under conditions of pathological metabolism. These mainly include diabetes mellitus and states of metabolic stress (e.g. intra-, and postoperatively, severe disease, injury), hormonally mediated depression of glucose tolerance, which can even lead to hyperglycaemia without exogenous supply of the substrate. Hyperglycaemia can – depending on its severity – lead to osmotically mediated renal fluid losses with consecutive hypertonic dehydration, to hyperosmotic disorders up to and including hyperosmotic coma.

Metabolism of glucose and electrolytes are closely related to each other. Insulin facilitates potassium influx into cells. Phosphate and magnesium are involved in the enzymatic reactions associated with glucose utilization. Potassium, phosphate and magnesium requirements may therefore increase following glucose administration and may therefore have to be monitored and supplemented according to individual needs. Especially cardiac and neurological functions may be impaired without supplementation.

Elimination

The final products of the complete oxidation of glucose are eliminated via the lungs (carbon dioxide) and the kidneys (water).

Practically no glucose is excreted renally by healthy persons. In pathological metabolic conditions associated with hyperglycaemia (e.g. diabetes mellitus, postaggression metabolism), glucose is also excreted via the kidneys (glucosuria) when the maximum tubular reabsorption capacity is exceeded (at blood glucose levels higher than 160–180 mg/dl or 8.8–9.9 mmol/l).

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Water for injections

6.2 Incompatibilities

Because glucose solutions have an acidic pH, incompatibilities can occur on mixing with other medicinal products and with blood. Information on compatibility can be requested from the manufacturer of the added drug.

Erythrocyte concentrates must not be suspended in glucose solutions because of the risk of pseudo-agglutination. See also section 4.4.

6.3 Shelf life

36 months

6.4 Special precautions for storage

Do not store above 25 °C.

6.5 Nature and contents of container

Bottles of colourless low-density polyethylene, contents: 500 ml available in packs of: 10 × 500 ml

6.6 Special precautions for disposal and other handling

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Containers are for single use only. Discard containers and any unused content after use.

Do not re-connect partially used containers.

Only to be used if the solution is clear and colourless or slightly yellowish and if the bottle and its closure are undamaged.

Administration should commence immediately after connecting the container to the giving set or infusion equipment.

Before addition of an additive or preparing a nutrient mixture, physical and chemical compatibility must be confirmed. Because glucose solutions have an acidic pH, incompatibilities can occur on mixing with other medicinal products. Information on compatibility can be requested from the manufacturer of the added drug.

When adding additives observe usual precautions of asepsis strictly.

7. DATE OF REVISION OF THE TEXT

Last revision: August 2023

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Product registration holder and manufactured by:
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